

Ball Balancing Robot Parts List

Location https://github.com/signag/Ball-Balancing-Robot_CS
Adapted from <https://github.com/I-M-Robotics-Lab/Ball-Balancing-Robot>
Model (Model.step) <https://makerworld.com/en/models/1197770-ball-balancing-robot>

Original Part Name	Quantity	Model Group	Model Part	Amazon	Remark
4-10 Bearing	6	Arm	B-4-10	https://amzn.eu/d/01DfXSLc	
SA5 Ball Joint	3	Arm	Ball_Joint_SA5	https://amzn.eu/d/0iQYb56e	Missing in original parts list
M3 x 10 Socket Head Screw	3	Arm	M3-10	https://amzn.eu/d/0g7pXmoE	
M3 x 15 Socket Head Screw	3	Arm	M3-15	https://amzn.eu/d/0efDuQfH	Used M3 x 16 from screws set
M3 x 15 Socket Head Screw	3	Arm	M3-15 v1.2, 4, ..	https://amzn.eu/d/0efDuQfH	Used M4 x 16 from screws set. A washer has been placed between arms
M4 Nylock Nut	3	Arm	Nylock-M4	https://amzn.eu/d/01R6Csu9	
Servo Arm	3	Arm	SERVO_ARM		Comes with Servo
M5x10x7 Threaded Inserts	3	Arm		https://amzn.eu/d/018Acx0K	Missing in original parts list
Rubber Foot 12x9x9	3	Assembly	12x9x9-M3	https://amzn.eu/d/026W2cM1	
M3 Hex Nut	3	Assembly	Hexnut-M3	https://amzn.eu/d/0dckAr4E	
M3 x 5 Socket Head Screw	6	Assembly	M3-5	https://amzn.eu/d/0efDuQfH	M3 x 5 too short. Took M3 x 12 from srews set
M3 x 5 Socket Head Screw	3	Assembly	M3-5	https://amzn.eu/d/0efDuQfH	Used M3 x 6 from srews set
M3 x 5 Socket Head Screw	6	Assembly	M3-5 (Mirror)	https://amzn.eu/d/0efDuQfH	Used M3 x 6 from srews set
M3 x 10 Standoff	3	Assembly	stand1	https://amzn.eu/d/077jvl7y	
M3 x 15 Standoff	3	Assembly	stand1	https://amzn.eu/d/077jvl7y	
M3 x 10 Standoff	6	Assembly	stand2	https://amzn.eu/d/077jvl7y	
M3 x 20 Standoff	6	Assembly	stand2	https://amzn.eu/d/077jvl7y	
Servo DS3218MG	3	Base	DS3218MG	https://amzn.eu/d/0aAt6Zfu	
M4 Hex Nut	12	Base / Fasteners	Hexnut-M4	https://amzn.eu/d/0efDuQfH	
M4 x 20 Socket Head Screw	6	Base / Fasteners	M4-20	https://amzn.eu/d/0aO0YwT4	M4 x 20 too short by ~3 mm. Used M4 x 30
M4 x 30 Socket Head Screw	6	Base / Fasteners	M4-30	https://amzn.eu/d/0aO0YwT4	
Barrel Jacks Female	1	Electronics	Barrel_Jacks_Female	https://amzn.eu/d/0f83lwsn	
PCA9685	1	Electronics	PCA9685	https://amzn.eu/d/0drExlMz	
PiCam V2	1	Electronics	PiCamV2.1	https://amzn.eu/d/083Phkyg	Used PiCam V3 Wide instead of V2
Raspberry Pi 3b plus	1	Electronics	raspberrypi_3_b_plus	https://amzn.eu/d/05NyQZgr	
M2.5 x 10 Cap Head Socket Screw	4	Electronics/Fasteners-Elec	M2.5-10	https://amzn.eu/d/0efDuQfH	M2.5 is too large. Used M2 x 8
M3 x 5 Socket Head Screw	9	top_ring	M3-5	https://amzn.eu/d/0efDuQfH	M3 x 5 was too short. Used M3 x 8 for 4 mm top panel
M5 x 30 Socket Head Screw	3	top_ring	M5-30	https://amzn.eu/d/01xLHwsY	
M4 Nylock Nut	3	top_ring	Nylock-M4	https://amzn.eu/d/00cSILYe	Must be Nylock M5
M5 Washer	12	top_ring		https://amzn.eu/d/0efDuQfH	
M2.5 x 4 x 3.5 Threaded Inserts	8	Electronics	Bottom	https://amzn.eu/d/018Acx0K	
M2.5 x 4 Socket Head Screw	8	Electronics	Bottom	https://amzn.eu/d/05jGSxSW	